

TABLE I
EEG SIGNAL FREQUENCY BANDS AND THEIR SIGNIFICANCE

Frequency Band	Frequency	Significance
Delta	1Hz-4Hz	Deep sleep
Theta	4Hz-8Hz	Creativity, dream sleep, drifting thoughts
Alpha	8Hz-12Hz	Relaxation, calmness, abstract thinking
Beta	12Hz-32Hz	Relaxed focus, high alertness, agitation, anxiety

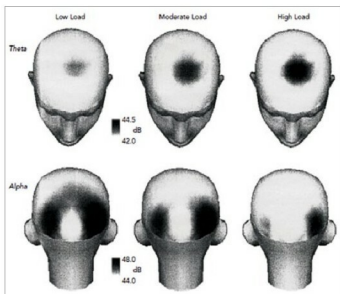


Fig. 5: Correlation of theta and alpha bands with complexity of task

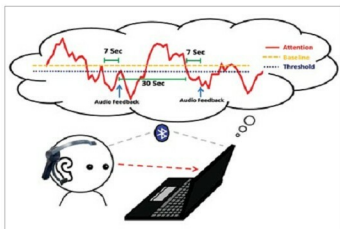


Fig. 6(a): Biofeedback system



Fig. 6(b): Single-channel EEG headset

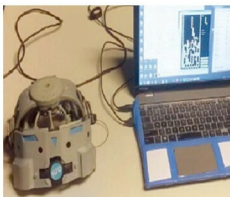


Fig. 7: 24-channel EEG headset

Researchers have proposed an engagement index, that is, Beta/(Alpha + Theta) to know the engagement of a learner while learning.

Since, EEG signals are very weak (in order of micro volts) and usually contain a lot of noise, these signals need to be amplified and filtered. To remove potential EEG artefacts, commonly, a band-pass filter of 3Hz to 100Hz is applied to the raw EEG signal. A notch filter can also be applied to eliminate electrical noises from the power source, which varies from 50Hz to 60Hz, depending on the geographical location.

Existing works across the world

Gevins et al. (2016) showed a grand average of topographical distribution of spectral power at peak

frequency in theta and alpha bands in each load level of the learner in the working memory task as shown in Fig. 5. It was observed that theta is larger in the moderate load and high load levels than in the low level, while alpha decreases monotonically from low level to high load levels as theta is related to creativity, while alpha is related to relaxation and calmness.

Sun and Yeh (2016) analysed the effects of attention monitoring with EEG biofeedback on university students' attention and self-efficacy by using anti-phishing instructional materials. Participants wore brainwave headsets when reading instructional materials, so that brainwave data could be collected and transferred to the monitoring system via Bluetooth. The system determined whether the participant's attention state fit the feedback threshold; once the threshold was reached, feedback on poor attention was provided to the participant as shown in Fig. 6(a). This system showed tremendous improvement in the learning of e-content.

They used NeuroSky MindSet, an audio headset equipped with a single-channel EEG sensor as shown in Fig. 6 (b). Unlike the multi-channel electrode nets worn in labs, this sensor requires no gel or saline for recording and therefore requires much less expertise to position. The scientists found that MindSet distinguished two fairly similar mental states (neutral and attentive) with 86 per cent accuracy.

Mills et al. (2017) detected cognitive load during learning by using QUASAR 24-channel EEG headset featuring dry electrodes that fits on the head similar to a hat, as shown in Fig. 7. It was designed to be a light, low-cost unit, specifically developed for ecological data collection. The system is self-contained, including data acquisition, data storage, cable and wireless data output, and batteries. The headset can be put on the head similar to wearing a baseball cap while reliably positioning the sensors. Signal quality recorded from the dry sensors showed to have a 90 per cent correlation to data obtained from clinical wet electrodes attached by a technician. Table II summarises the work of researchers in a similar direction.